

The Genesis of Relativistic Mechanics: Time, the Speed of Light, and Gravity as Emergent Properties of Baryonic Impedance on the Planck Frequency

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Abstract

This paper proposes a deterministic mechanical origin for time, the speed of light (c), and gravity, demonstrating that none are fundamental dimensions or axiomatic constants. By redefining the primordial spatial vacuum as a continuous wave resonance constrained by the Planck frequency (f_p), we establish a timeless baseline state. We theorize that the absolute baryonic mass of the universe (1.53×10^{53} kg) functions as an absolute topological impedance, generating a localized frequency drop governed by a universal time-scaling ratio ($S_r \approx 7.03 \times 10^{60}$). Within this framework, time emerges exclusively from the latency differential between the absolute Planck resonance and the baryon-dampened local frequency. Consequently, the speed of light is mathematically derived not as a predetermined speed limit, but as the exact geometric translation of Planck lengths parsed per single impeded baryonic pulse. Furthermore, gravity is redefined entirely: it is the inherent thermodynamic drive of baryonic matter to return to primordial singularity, mathematically bounded and counteracted by the outward

pressure of the expanding spatial wave resonance. The exact intersection of these two forces dictates the macroscopic structural limits of the universe.

1. Introduction

The standard Λ CDM cosmological paradigm interprets time, the speed of light (c), and gravity as axiomatic foundation blocks of the universe — parameters that simply exist without a mechanical origin. However, recent formulations of the Phantom Metric [1] demonstrate that macroscopic spatial anomalies, such as flat galactic rotation curves and excessive gravitational lensing, emerge deterministically from macroscopic boundary conditions rather than invisible particulate mass. Building upon this geometric architecture, this paper performs a mechanical reverse-engineering of the fundamental constants. We propose that spacetime is not a unified continuum; instead, spatial geometry strictly precedes time. By defining the primordial vacuum as a continuous kinematic resonance, we demonstrate that macroscopic time and gravity are strictly emergent thermodynamic properties—the direct physical manifestations of baryonic mass functioning as a topological impedance against the absolute fundamental frequency of space.

2. The Pre-Temporal State: Absolute Spatial Resonance

In standard cosmology, space and time are inextricably linked as spacetime, originating simultaneously at the Big Bang. This framework separates the two, proposing that spatial geometry precedes time. We define the primordial substrate not as an expanding balloon, but as a continuous field of geometric resonance operating at the maximum theoretical limit: the Planck frequency ($f_P \approx 1.85 \times 10^{43}$ Hz). Each pulse constitutes one Planck length (l_P). Crucially, in the absence of baryonic mass, this resonance is absolute. Without a slower reference

point to establish relativity, the concept of a "second" does not exist. The pre-temporal universe is a state of continuous kinematic flux, yet strictly timeless.

3. The Introduction of Baryonic Impedance and the Emergence of Time

Time is conventionally treated as a flowing dimension. We propose that time is strictly an emergent property of resistance.

When the total baryonic mass of the universe ($M_{tot} \approx 1.53 \times 10^{53}$ kg) [2] was introduced into the expanding spatial resonance, it acted as a mechanical impedance. Baryonic matter cannot oscillate at the Planck frequency; it absorbs and dampens the local spatial pulses, creating a forced resonance lag.

Drawing from our macroscopic framework, this mechanical resistance generates a Universal Time-Scaling Ratio ($S_r \approx 7.03 \times 10^{60}$) [2]. This ratio is the explicit topological load that slows the primordial metric. Time (t) is born exactly at the intersection of these two rates. It is the latency differential between the absolute spatial pulse (f_P) and the burdened frequency of the baryonic mass ($f_{baryonic}$):

$$\Delta f = f_P - f_{baryonic}$$

Where $f_{baryonic}$ is derived by dividing f_P by the baryonic impedance factor S_r . Macroscopic time, therefore, is not a dimension objects travel through; it is the physical sensation of frictional drag against the absolute resonance of the universe.

The concept of time is strictly defined by this periodicity. A standard "second" is simply the duration required for the unhindered spatial metric to complete 1.85×10^{43} pure Planck cycles. Conversely, due to the extreme impedance of the total baryonic mass, a single complete macroscopic cycle of the loaded metric requires approximately 3.8×10^{17} seconds. The ratio between these absolute spatial cycles and the burdened macroscopic period is exactly S_r .

4. Deriving the Speed of Light (c) from Temporal Latency

If time is defined by the latency differential (Δf), then velocity is restricted by it. The speed of light (c) is universally accepted as an arbitrary constant. This framework derives c deterministically as a geometric product.

A photon, possessing zero resting mass, bypasses the baryonic impedance factor (S_r) entirely. It rides the absolute spatial resonance, advancing exactly one Planck length (l_p) per Planck pulse (f_p). The maximum causal velocity of the spatial metric is therefore strictly defined at the quantum scale:

$$c = l_p f_p$$

However, velocity inherently requires a temporal denominator (distance/time). Because human observation is constrained by baryonic time, we measure the photon's progress relative to our slowed reference frame. The speed of light ($c \approx 299,792,458$ m/s) is therefore not an arbitrary cosmic speed limit; it is the deterministic arithmetic product of absolute spatial gaps (l_p) parsed during one unit of impeded baryonic time. It is the exact translation of pure, timeless spatial geometry forced into a delayed temporal denominator.

5. Gravity as the Thermodynamic Drive Toward Singularity and Spatial Tension

Under General Relativity, gravity is the curvature of spacetime. Under the Standard Model, it is an undetected quantum force. This framework posits that gravity is neither.

If the spatial resonance acts as an outward, expanding pressure wave, baryonic matter exists in a state of unnatural geometric dispersion. Gravity is the inherent

thermodynamic memory of baryonic matter — the constant, mechanical drive to collapse back into its original primordial singularity (zero entropy). Masses do not "attract" each other through empty space via force carriers (gravitons). Instead, gravity is the passive collapse of matter toward the lowest energy state, characterized by the Newtonian inward acceleration:

$$a_g = \frac{GM}{r^2}$$

However, this thermodynamic collapse is actively counteracted by the outward pressure of the spatial wave resonance. As derived from our macroscopic geometric baseline [2], the continuous expansion of the spatial metric exerts a constant topological tension (a_τ) on the fabric of space:

$$a_\tau = cH_0$$

The macroscopic architecture of the universe is governed by a perpetual struggle between these two states. Close to a mass center, baryonic matter successfully bends space, striving to cluster back into singularity. However, as radial distance increases, the matter's gravitational "grip" weakens. When the thermodynamic drive for singularity drops exactly to the level of the background topological tension ($\frac{GM}{r^2} = cH_0$), a critical geometric phase transition occurs.

At this exact boundary, the radial distance (r) loses all kinematic authority. The mass can no longer pull the outer baryonic material inward, as the outward topological tension (a_τ) overrides the local collapse. The system mathematically locks into a constant geometric state. The empirical proofs for this exact saturation boundary are comprehensively demonstrated in the Phantom Metric framework [1, 3], dictating both flat galactic rotation curves and the macroscopic boundaries of strong gravitational lenses without the need for dark matter.

6. Conclusion

By reverse-engineering the fundamental constants, we eliminate the theoretical necessity for cosmic inflation, dark matter, and quantum gravity paradoxes. Spacetime is not a unified dimension; space is a continuous resonance, and time is the macroscopic latency generated by baryonic impedance (S_r). The measurable velocity of light (c) is not an independent parameter, but simply absolute spatial geometry ($l_P f_P$) projected through this delayed temporal denominator. Furthermore, gravity is the thermodynamic tension between primordial singularity and spatial expansion.

References

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